

EXAM 7 – SPRING 2015

4. (3.5 points)

Given the following information:

Age-to-Age Paid Loss Development Factors

Accident Year	12-24 Months	24-36 Months	36-48 Months	48-60 Months	60-72 Months	72-84 Months
2008	7.00	2.55	1.60	1.30	1.20	1.05
2009	6.00	2.60	1.40	1.50	1.10	
2010	5.00	2.50	1.80	1.20		
2011	5.50	2.80	1.50			
2012	8.00	2.40				
2013	7.00					

- z-value for the 95% confidence level: ± 1.96

a. (3 points)

Using a 95% confidence threshold, test the null hypothesis that the triangle does not display significant calendar year effects.

b. (0.5 point)

Provide two examples of calendar years effects that may cause loss development data not to be independent by accident year.

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QUESTION 4

TOTAL POINT VALUE: 3.5

LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as chain ladder, Cape Cod, chain ladder plus calendar-year effects, and Bornhuetter-Ferguson

SAMPLE ANSWERS

Part a: 3 points

Sample Answer 1

AY	12-24 Months	24-36 Months	36-48 Months	48-60 Months	60-72 Months	72-84 Months
08	L	*	L	*	L	*
09	S	L	S	L	S	
10	S	S	L	S		
11	S	L	S			
12	L	S				
13	L					
Median	6.50	2.55	1.55	1.30	1.15	1.05

S indicates less than median for development age

L indicates greater than median for development age

* indicates equal to median for development age

A_j = diagonal j

S_j = # of S in diagonal J; L_j = # of L in diagonal J

N_j = S_j + L_j

M_j = (n-1)/2; round down

Z_j = Min (S_j, L_j)

Ignore j=1 since only one element

j	S	L	n	m	Z	E[Zn]	Var[Zn]
2	1	0	0	0	0	0	0
3	1	2	3	1	1	0.750	0.188
4	3	0	3	1	0	0.750	0.188
5	0	5	5	2	0	1.563	0.371
6	4	1	5	2	1	1.563	0.371
Total					Z = 2	4.625	1.117

n=1 E(z)= 0 Var(z)=0

n=3

z	Comb	Prob
0	2	2/(2+6)=0.25
1	3*2=6	6/(2+6)=0.75

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$$E(z)=0.25(0)+0.75*(1)=0.75$$

$$E(z^2)=0.25(0)^2+0.75*(1)^2=0.75$$

$$\text{Var}(z)=0.75 - 0.75^2=0.1875$$

n=5

Z	Comb	Prob
0	2	2/32
1	5*2=10	10/32
2	20	20/32

$$E(z)=2/32(0)+10/32(1)+20/32(2)=1.5625$$

$$E(z^2)=2/32(0)^2+10/32(1)^2+20/32(2)^2=2.8125$$

$$\text{Var}(z)=2.8125 - (1.5625)^2=0.371$$

$$Z=\text{Sum}(Z_j)=2$$

$$E(Z)=\text{Sum}(E(Z_j))=4.625$$

$$\text{Var}(Z)=1.117$$

$$95\% \text{ CI } 4.625 \pm 1.96 * \text{Sqrt}(1.117)$$

$$(2.554, 6.696)$$

Z=2 is not inside CI. So reject H0 that there are no CY effects.

Sample Answer 2

08	L	*	L	*	L	*
09	S	L	S	L	S	
10	S	S	L	S		
11	S	L	S			
12	L	S				
13	L					

For j=2, S=1, L=0, Z=min(S,L)=0, n=S+L=1, M=[(n-1)/2]=0, $E(Z)=1/2 - \text{Comb}(1-1,0)*1/2=0$

$$\text{Va}(Z)r=1/2*(1-1)-\text{Comb}(1-1,0)*1/2*(1-1)+0-0^2=0$$

For j=3, S=1, L=2, Z=1, n=3, M=1, $E(Z)=3/2 - \text{Comb}(3-1,0)*3/2^3=.75$

$$\text{Var}(Z)=3/2*(3-1)/2 - \text{Comb}(3-1,1)*3/2^3*(3-1)+.75-.75^2=.1875$$

For j=4, S=3, L=0, Z=0, n=3, M=1, $E(Z) = .75$, $\text{Var}(Z)=.1875$

For j=5, S=0, L=5, Z=0, n=5, M=2, $E(Z)=5/2 - \text{Comb}(5-1,2)*5/2^5=1.5625$

$$\text{Var}(Z)=5/2*(5-1)/2 - \text{Comb}(5-1,2)*5/2^5*(5-1)+1.5625-.1.5625^2=0.371$$

For j=6, S=4, L=1, Z=1, n=5, M=2, $E(Z) = 1.5625$, $\text{Var}(Z)=.371$

$$Z=0+1+0+0+1=2$$

$$E(Z)=0+0.75+0.75+1.5625+1.5625=4.625$$

$$\text{Var}(Z)=0+0.1875+0.1875+0.371+0.371=1.117$$

$$4.625-1.96*1.117^{.5}=2.554$$

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$Z=2<2.554$ Reject the null hypothesis – The triangle displays significant calendar year effect
Part b: 0.5 point
<i>Sample Answer 1</i> • Claims department process change could cause a strengthening of reserves for all AYs leading to an unusually strong diagonal • A court ruling with impact on claims that already occurred could cause all AY's to shift during a calendar year <i>Sample Answer 2</i> • A change in claims handling system can affect calendar year claims development • A legislative change affecting benefit levels can also affect CY claims because it applies to claims from all AYs <i>Other responses that made mention of any of the following were accepted as one of the two responses required:</i> • High inflation • Changing inflation • Changes in payment processes
EXAMINERS' REPORT
The topic tested is clearly identified on the syllabus and the exam problem was very similar to the example in the Mack paper. In general, candidates did well on this question; about a third of the candidates earned full credit.
Part a • Most candidates did not show the calculations for the median LDFs for each evaluation. However, no deduction was made if the 'rank' picture was correct. • A number of candidates solved the problem using Spearman's T Method. However, this did not receive credit because the method is a development year test while the problem was looking at calendar year effects.
Part b A common response was the single word 'inflation'. This did not receive credit because it is changes in inflation that cause calendar year effects. However, 'high inflation' was accepted because it implied that inflation was increasing.